

Simulation Platform for an Early 20th-Century Domestic Energy Management Apparatus

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Abstract

This paper presents the design of a digital twin for a domestic electrical energy system originally installed in a rural estate in Chamusca (Portugal) around 1920 and donated in 2023 to the Faraday Museum at Instituto Superior Técnico (IST). The system integrates a Crossley internal combustion engine, an ASEA direct-current dynamo, an ASEA three-phase induction motor, and a Tudor open-cell lead-acid battery bank. Based on technical documentation produced by the Faraday Museum during the system’s dismantling and restoration, this work outlines a digital preservation platform combining three components: (1) a 3D reconstruction in Blender, (2) an interactive real-time simulation in Unity, and (3) a Retrieval-Augmented Generation (RAG) system for accessing historical data. The project aims to preserve the technological heritage of early domestic electrification in Portugal and provide an educational tool for researchers and museum visitors.

Keywords: Digital cultural heritage; Domestic electricity; Digital twin; Simulation; Lead-acid batteries; Portugal

Highlights

- Documentation of a rare 1920s domestic hybrid energy system (Thermal/Electric).
- Development of a digital twin using Blender (modelling) and Unity (simulation).
- Integration of RAG technology to query historical museum archives.
- Collaboration with Museu Faraday (IST) for technical validation and restoration context.

1 The Dawn of Electricity in Portugal

The use of electricity has been one of the greatest technological achievements of humanity and Portugal was among the first countries to embrace it. This development can be traced back to the second half of the 19th century when, in 1878, the Belgian electrical engineer Zénobe Théophile Gram

and his team constructed an alternator capable of powering eight Yablochkov candles (sometimes an electric candle or a Jablochkoff candle) Walker (1878); 653 (1924). These candles were used to illuminate the Avenue de l’Opéra and the Place de l’Opéra during the Paris World Fair Exposition of 1878 Guarnieri (2015). The system was first switched on in May, marking the first large-scale public demonstrations of electric lighting.

Later that same year, in September, the town of Cascais became the stage for the very first experience of public electric lighting in Portugal. On the night of September 28, 1878, marking the 15th birthday of Prince D. Carlos, the Cascais Citadel was illuminated with six electric lamps brought directly from Paris. These candles projected an intense, white light ideal for large public spaces—caused a sensation among those present and caught the attention of the international press.

The history of domestic electrification in Portugal typically focuses on large infrastructure, but surviving examples of autonomous domestic generation systems are rare. This paper focuses on a specific apparatus: a domestic energy management system installed c. 1920 in a rural es-

tate (*quinta*) at Rua Direita de São Pedro 194, Chamusca (Piedade, 2023b). This system provided local generation, storage, and distribution of electricity for lighting and agricultural machinery before the widespread arrival of the public grid.

In March 2023, the system was donated by its owner, Eng. Miguel Tavares Pestana, to the Faraday Museum at Instituto Superior Técnico (IST). Following the donation, a team led by Moisés Piedade conducted a detailed identification, dismantling, and transport operation to preserve the equipment (Piedade, 2023b).

This article describes the development of a simulation platform designed to act as a “digital twin” of this historical apparatus. The project addresses the need to make this complex machinery accessible to the public and researchers, as physical operation of the restored artifacts may not always be feasible. The platform integrates high-fidelity 3D modelling, physics-based simulation of electrical components, and AI-assisted documentation retrieval.

2 Historical and Technical Background

2.1 Provenance and Historical Context

The installation was originally located in a *quinta* belonging to the family of Eng. José de Mascarenhas Pedroso Belard da Fonseca (1889–1969), a prominent civil engineer and former director of IST (1942–1953) (Piedade, 2023b). The estate included agricultural infrastructure for cereal threshing and olive oil production, powered by steam and later by the internal combustion system described here. The energy system was housed in a dedicated machine room adjacent to a water well (*mina*) and reservoir.

2.2 System Architecture and Evolution

Technical analysis performed by the Faraday Museum team indicates the system evolved in three distinct phases (Piedade, 2023b,a):

1. **Mechanical Phase (pre-1918):** A Crossley single-cylinder internal combustion engine (estimated 5–10 hp) was installed to drive water pumps for irrigation, replacing an animal-driven water wheel (*nora*).
2. **DC Electrification Phase (c. 1918–1923):**

- In May 1918, a dynamo (DC generator) manufactured by ASEA (approx. 3 kW) was coupled to the engine.
- Between 1921 and 1923, a bank of 60 open-cell lead-acid batteries (Tudor type L1, No. 969) was installed to provide energy storage and 24-hour availability.

3. **AC Integration Phase (c. 1929):** Following the arrival of the public grid in Chamusca, an ASEA three-phase induction motor was added (c. 1929) to drive the dynamo, relegating the Crossley engine to mechanical pumping duties.

2.3 Component Specifications

The physical system currently preserved at the Faraday Museum comprises five main components:

- **Thermal Engine:** Crossley single-cylinder, spark ignition via magneto, with two large flywheels (> 500 kg each).
- **Dynamo:** ASEA DC generator, 115 V–160 V, belt-driven.
- **Electric Motor:** ASEA 3-phase induction motor, 3 hp, 380 V/120 V (Star connection).
- **Battery Bank:** Tudor lead-acid, 60 cells, open glass container type.
- **Control Board:** Marble panel containing voltmeters, ammeters, a main rheostat, and a double-selector switch for regulating charge/discharge voltage across the final 20 cells.

3 Materials and Methods

3.1 Source Data and Documentation

The primary data source for the digital twin is the technical documentation produced by Piedade (2023b) and Piedade (2023a) during the 2023 recovery operation. This includes:

- Inventories of all mechanical and electrical components.
- Photographs of the original *in situ* arrangement.
- Transcriptions of equipment nameplates (dates, serial numbers, electrical ratings).

- Schematic diagrams of the DC distribution and battery regulation topology.

3.2 Digital Twin Architecture

The proposed platform consists of three integrated modules:

3.2.1 3D Modelling (Blender)

A 3D reconstruction of the machine room is being manually developed in Blender. The modelling and texturing process relies on visual references, including photographs and video footage, to faithfully recreate the geometry and surface details of the engine, dynamo, and switchboards.

- **[TODO]:** Insert specific details on polygon count or texture resolution once modelling is advanced.
- **[TODO]:** Add figure showing wireframe vs. rendered view of the Crossley engine.

3.2.2 Interactive Simulation (Unity)

The runtime environment is built in Unity. It simulates the flow of energy between components based on the logical states defined in the historical reports:

- **Logic:** State machine handling the transition between Engine-Drive and Motor-Drive modes.
- **Physics:** Simplified electrical model calculating Voltage (V) and Current (I) based on the load and the position of the rheostat and battery cell selectors.

3.2.3 RAG Documentation System

To support historical research, a Retrieval-Augmented Generation (RAG) system is integrated. It indexes the PDF reports from the Faraday Museum.

- **[TODO]:** Specifics of the vector database and LLM used will be added here upon implementation.

4 Results

[Note: This section is currently a placeholder for future simulation data.]

4.1 Visual Reconstruction

[TODO]: This section will present renderings of the restored digital apparatus.

4.2 Simulation Scenarios

The platform allows users to simulate specific historical scenarios documented by Piedade (2023a):

1. **Scenario A:** Running the house on batteries (Engine OFF).
2. **Scenario B:** Charging batteries using the Crossley engine (c. 1923).
3. **Scenario C:** Charging batteries using the AC Grid and Induction Motor (c. 1930).

5 Discussion and Conclusions

The Chamusca domestic energy system represents a transition period in Portuguese industrial history, moving from mechanical steam/combustion power to electrical grid integration. This project creates a permanent digital record of the installation. By combining the rigorous historical data provided by the Faraday Museum (Piedade, 2023b) with modern game-engine technology, the platform offers a novel method for disseminating technical heritage.

Future work includes finalizing the Unity simulation logic and deploying the system for public demonstration at IST.

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Declaration of Generative AI Use

During the preparation of this work, the author used [Insert Tool Name] to assist in [specific task, e.g., code structure generation]. The author reviewed and edited the content and takes full responsibility for the published article.

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